ENCN205

Applied Data Analysis for Civil and Natural Systems

LECTURE 10

Risk Assessment & Multi-Criteria Evaluation

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Think About It

"Risk = Hazard × Exposure × Vulnerability"

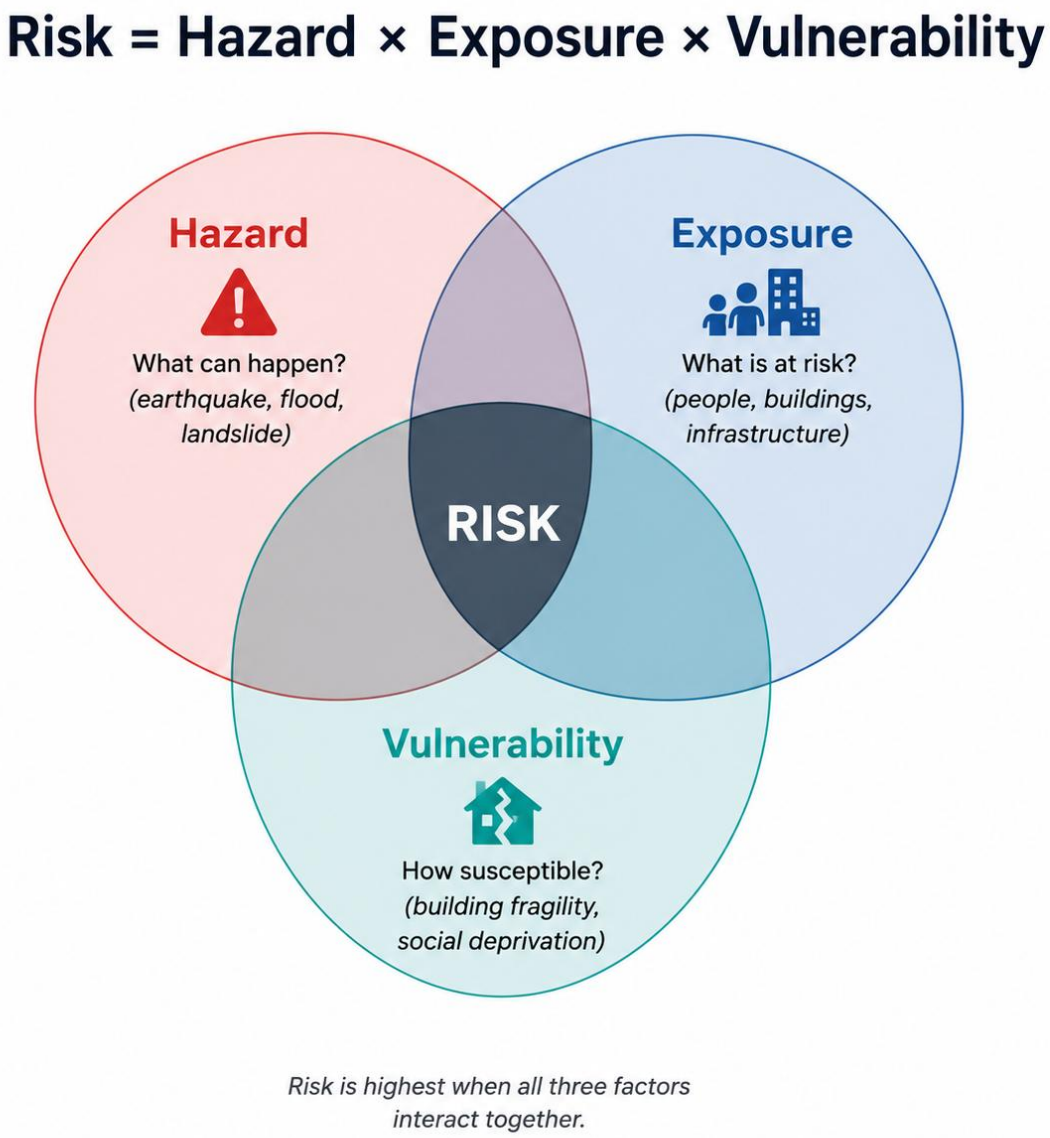
A magnitude 7 earthquake hits an uninhabited desert — high risk?

A magnitude 5 earthquake hits a dense city of old buildings — high risk?

A flood reaches a modern hospital with emergency generators — high risk?

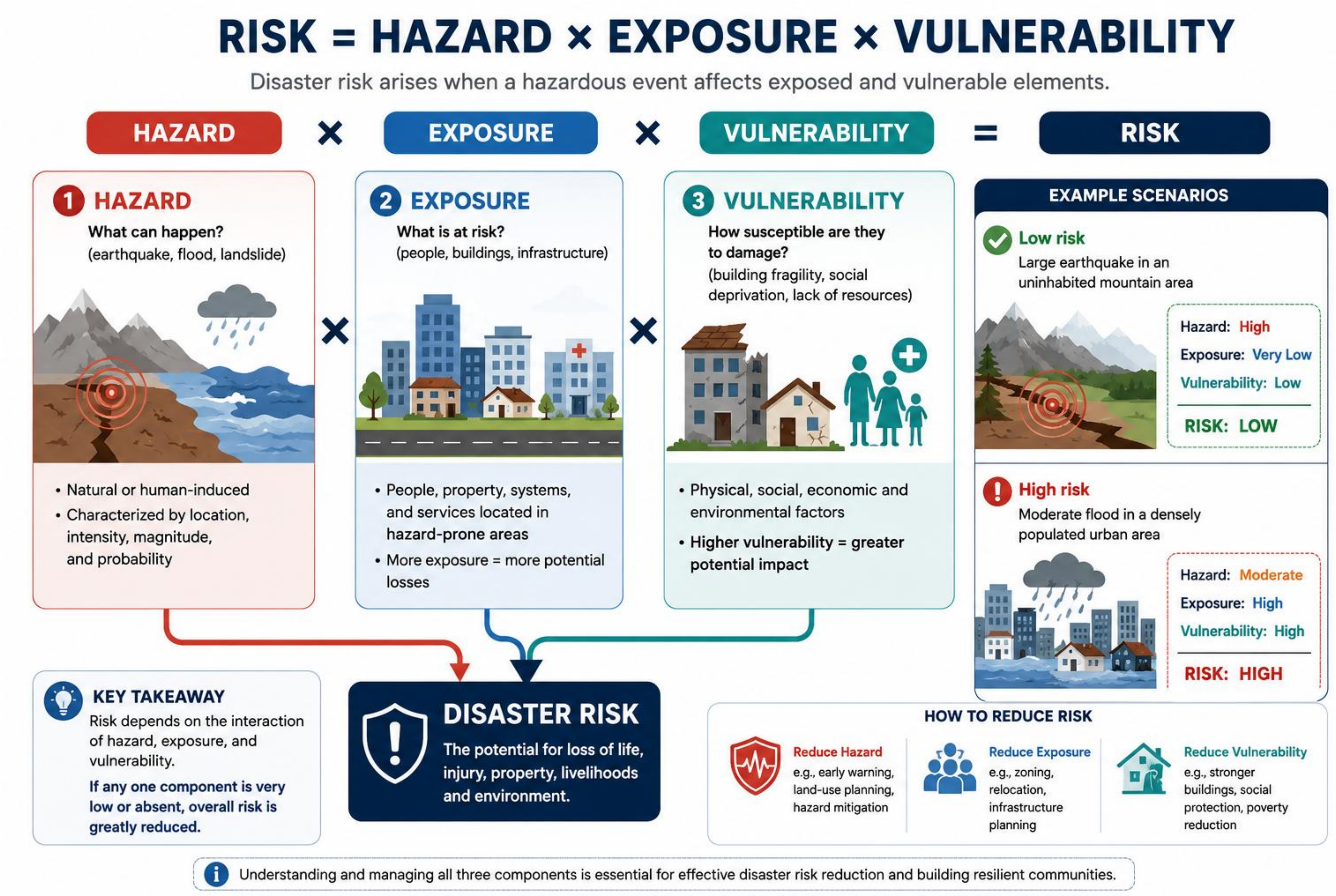
Risk depends on all three factors. GIS is the tool that overlays them.

The Risk Framework



Risk exists only where all three circles overlap: hazard is present, something is exposed, and it is vulnerable.

GIS overlays hazard maps + exposure data + vulnerability information to identify risk hotspots.


RISK = HAZARD × EXPOSURE × VULNERABILITY
Disaster risk arises when a hazardous event affects exposed and vulnerable elements.
HAZARD × EXPOSURE × VULNERABILITY = RISK
1 HAZARD
What can happen?
(earthquake, flood, landslide)
• Natural or human-induced
• Characterized by location, intensity, magnitude, and probability
2 EXPOSURE
What is at risk?
(people, buildings, infrastructure)
• People, property, systems, and services located in hazard-prone areas
• More exposure = more potential losses
3 VULNERABILITY
How susceptible are they to damage?
(building fragility, social deprivation, lack of resources)
• Physical, social, economic and environmental factors
• Higher vulnerability = greater potential impact
EXAMPLE SCENARIOS
Low risk
Large earthquake in an uninhabited mountain area
Hazard: High
Exposure: Very Low
Vulnerability: Low
RISK: LOW
High risk
Moderate flood in a densely populated urban area
Hazard: Moderate
Exposure: High
Vulnerability: High
RISK: HIGH
KEY TAKEAWAY
Risk depends on the interaction of hazard, exposure, and vulnerability.
If any one component is very low or absent, overall risk is greatly reduced.
DISASTER RISK
The potential for loss of life, injury, property, livelihoods and environment.
HOW TO REDUCE RISK
Reduce Hazard
e.g., early warning, land-use planning, hazard mitigation
Reduce Exposure
e.g., zoning, relocation, infrastructure planning
Reduce Vulnerability
e.g., stronger buildings, social protection, poverty reduction
Understanding and managing all three components is essential for effective disaster risk reduction and building resilient communities.

Mapping Hazards

Hazards are characterised by type, intensity, and return period:

Earthquakes — PGA maps, fault proximity, liquefaction susceptibility

Flooding — depth maps, extent maps, return period (e.g. 1-in-100 year)

Landslides — slope, geology, rainfall triggers

Tsunami — inundation zones, wave height, arrival time

Sea level rise — projected coastal inundation for +0.5m, +1m, +1.5m

Return period: a 1-in-100 year flood has a 1% probability per year.

In 50 years, the probability of at least one occurrence = 39%.

NZ sources: GNS Active Faults, NIWA flood models, LINZ coastal data.

Always ask: 'what return period — and how old is the model?'

Exposure — What Is at Risk?

Quantifying what lies within the hazard zone:

People — population density (census meshblocks), time-of-day patterns

Buildings — footprints, replacement value, occupancy type

Infrastructure — roads, bridges, pipes, power lines

Economic activity — business locations, agricultural land

Environment — ecosystems, cultural heritage sites

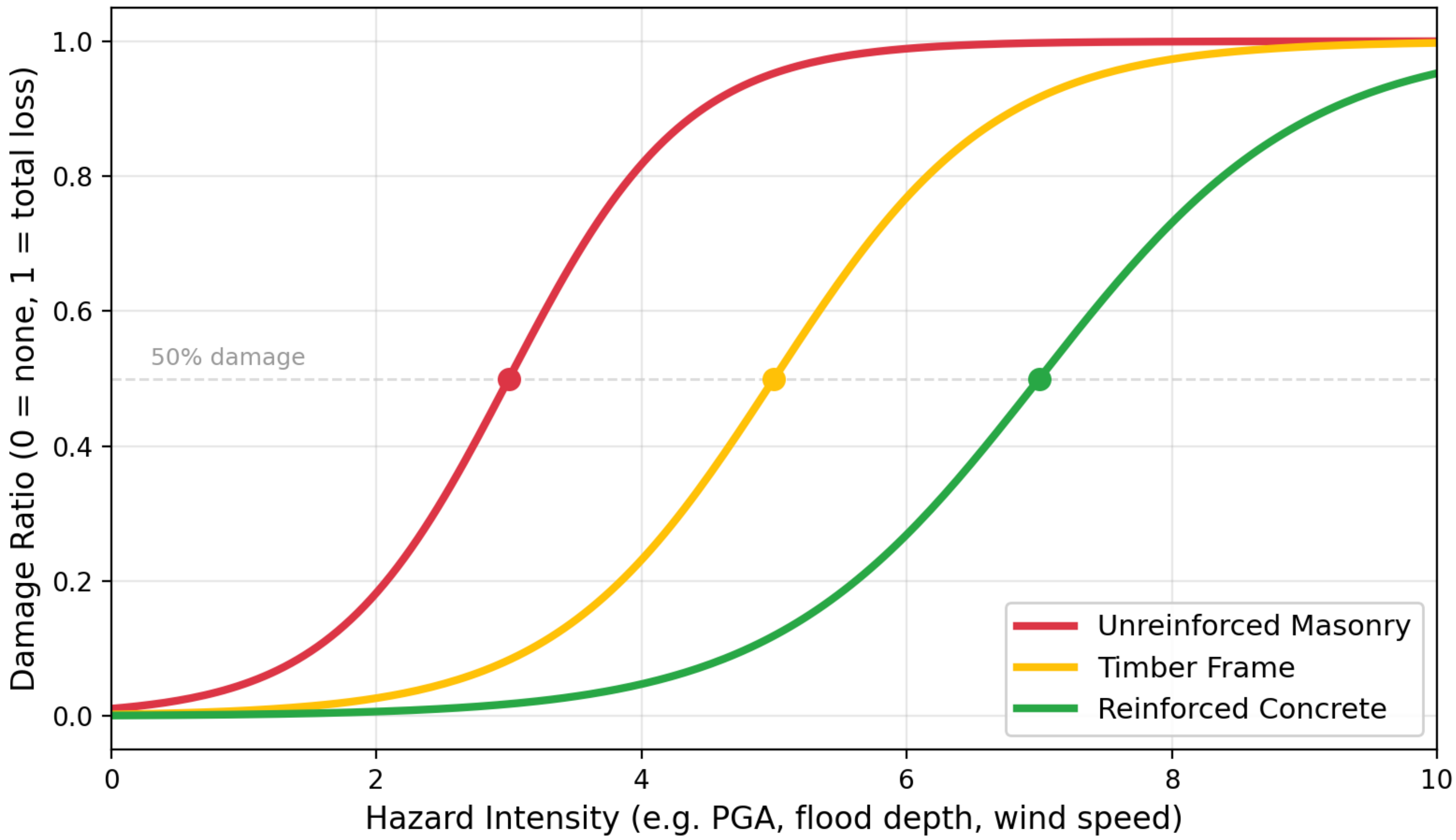
GIS operation: spatial join / overlay of hazard layer with asset layers.

Example: How many buildings are in the 1-in-100 year flood zone?

```
buildings_in_flood = gpd.sjoin(buildings, flood_zone_100yr)
```

Always ask: 'how current is the asset data?'

Vulnerability (Fragility) Curves — Physical Damage vs Hazard Intensity



Different building types sustain different damage at the same hazard intensity.

Vulnerability curves convert hazard intensity to expected damage.

Social Vulnerability

Physical vulnerability is not the whole picture:

NZ Deprivation Index (NZDep):

Composite of 9 census variables: income, employment, education, housing, etc.

Scale 1 (least deprived) to 10 (most deprived)

Available as meshblock-level spatial data

Why it matters for risk:

Deprived communities have less capacity to prepare, respond, and recover

Same hazard, same building stock → worse outcomes in deprived areas

Insurance rates lower, savings lower, recovery slower

GIS overlay: hazard + NZDep = identify communities at highest compound risk.

Consecuence Assessment -- Four Dimensions

1. Life Safety

Fatalities, injuries, displaced people

2. Economic

Direct damage costs, business interruption, repair

3. Social

Community disruption, cultural loss, wellbeing

4. Environmental

Ecosystem damage, contamination, biodiversity loss

GIS quantifies consequences spatially:

Overlay hazard + exposure + vulnerability

Sum damage across all exposed assets

Map consequence intensity spatially

Aggregate by zone (suburb, district, region)

Canterbury earthquakes:

185 fatalities

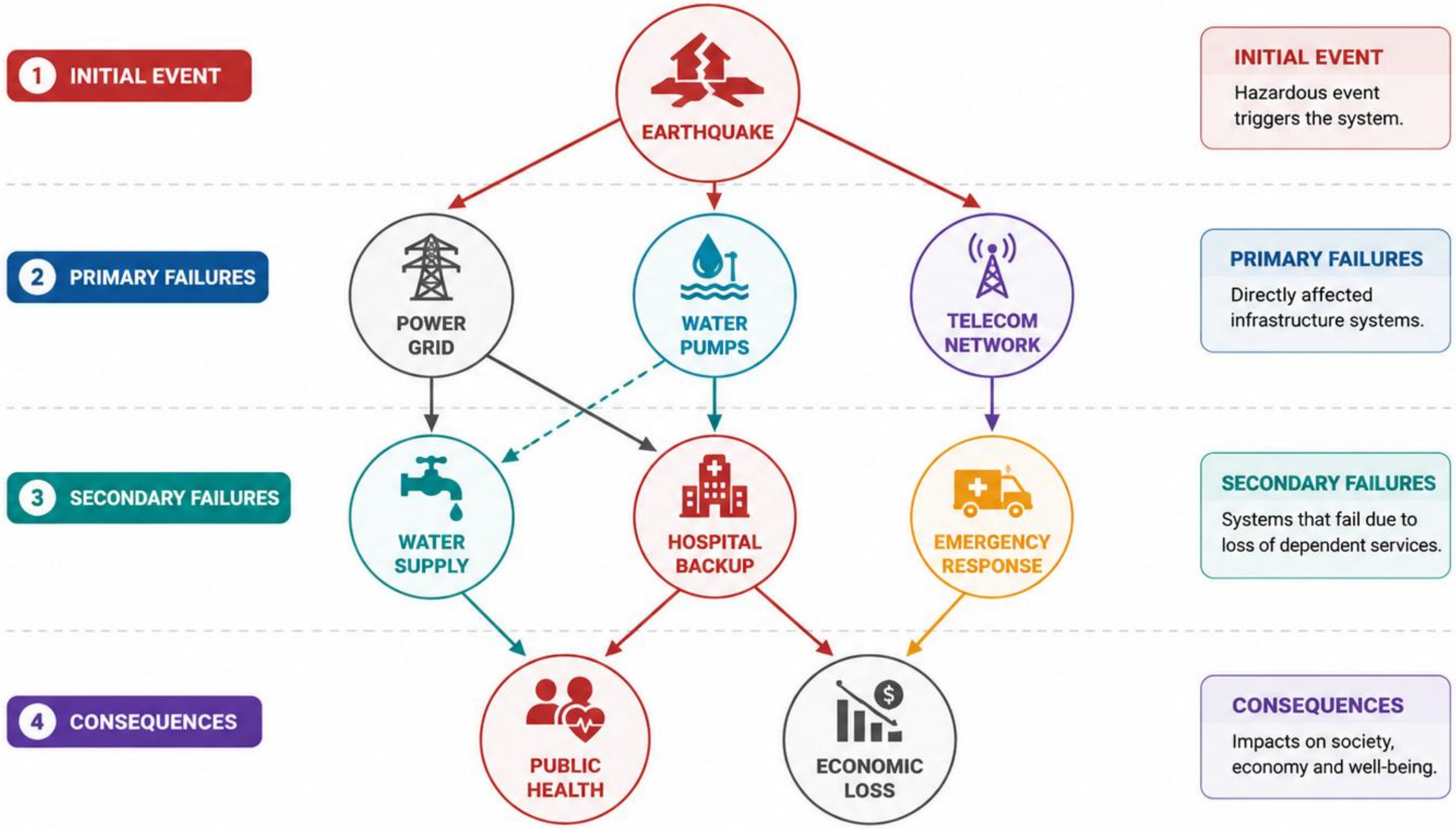
$40 billion economic loss

Massive social disruption

Avon River contamination

CASCADING INFRASTRUCTURE FAILURES – INTERDEPENDENCY NETWORK

Failure of one system can trigger failures in others, leading to severe consequences.



HOW TO READ

Arrows show the direction of impact or dependency.

→ **Direct dependency** Immediate impact between systems.

- - → **Indirect dependency** Impact occurs through another system.

KEY TAKEAWAY

Strengthening interdependencies and building redundancy reduces cascading failures.

Infrastructure systems are interdependent — failure in one can cascade through others.

Cascading Failures: Christchurch, 22 February 2011

The cascade, step by step

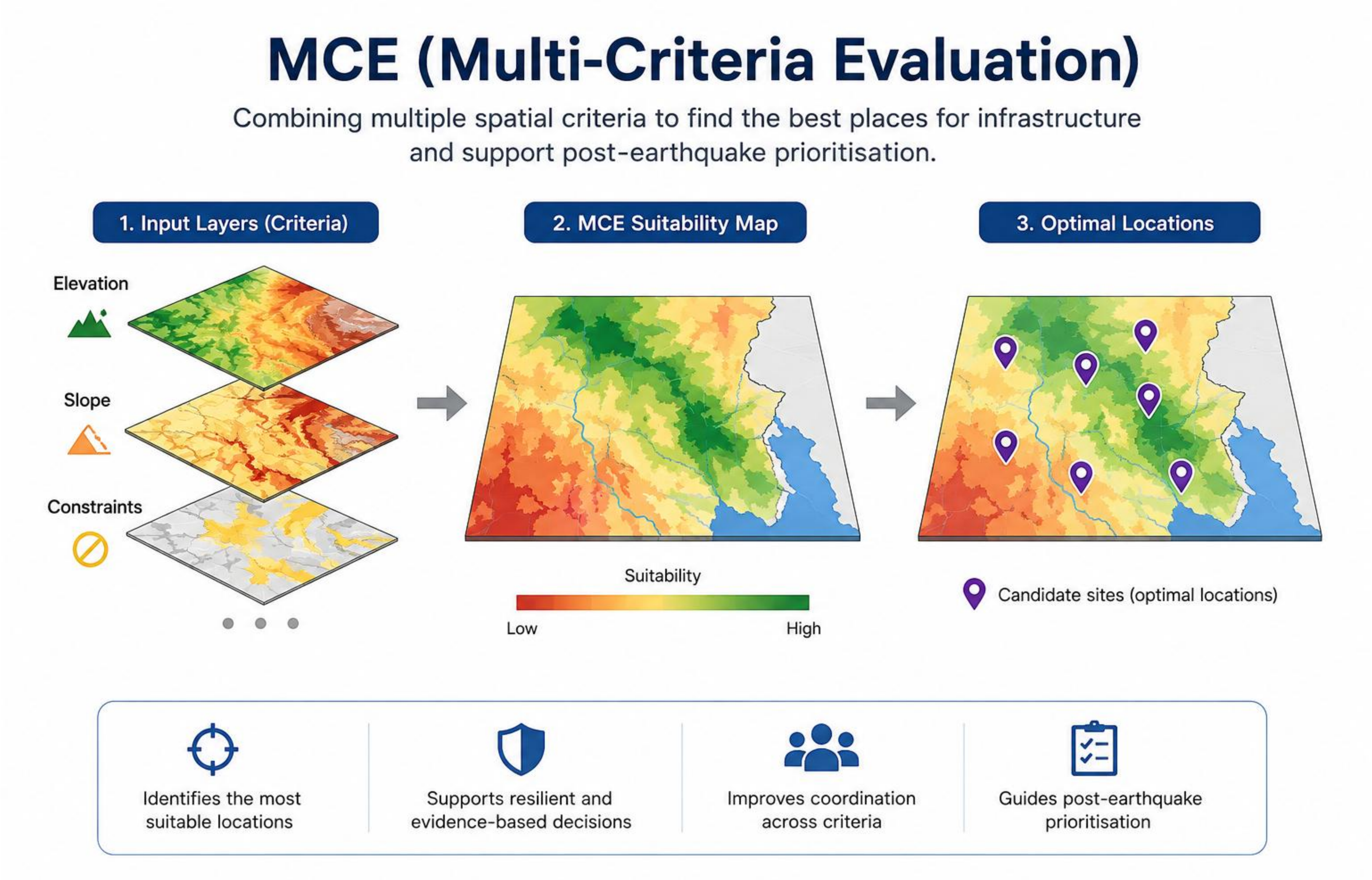
- Trigger: the M6.2 earthquake damaged electricity substations
- Power loss stopped water-supply and wastewater pumping stations
- No water pressure → hospitals and firefighting compromised; sewage overflowed into rivers
- Road damage and telecom outages slowed the repair crews — extending every other failure

One damaged system is an incident. Interdependence turns it into a crisis.

How GIS finds the weak points

- Map every asset AND what it depends on — a spatial dependency network
- Overlay hazard zones on the network: which dependencies sit in the same hazard area?
- Trace failures downstream: which single assets, if lost, cascade furthest? (e.g. a substation feeding 20 pumping stations)
- Prioritise those critical nodes for strengthening or redundancy

GIS makes the dependencies visible — before the earthquake does.



MCE maps multiple spatial criteria to identify optimal infrastructure siting. The same approach applies to post-earthquake infrastructure prioritization.

LIVE DEMO

L10_RiskAssessment.html

Adjust criteria weights interactively and see how the
optimal site changes — revealing the sensitivity of MCE to weight choices.

Scan to open the demo on your phone

aardid.github.io/uc_205_gis_course/L10_RiskAssessment.html

Multi-Criteria Evaluation (MCE) Workflow

1. Identify Criteria

Decide what factors matter.

→

2. Standardise Scores (0–1)

Convert all criteria to a common 0–1 scale.

→

3. Assign Weights

Set how important each criterion is relative to others.

→

4. Compute Composite

Calculate the weighted sum to get the overall score.

→

5. Sensitivity Analysis

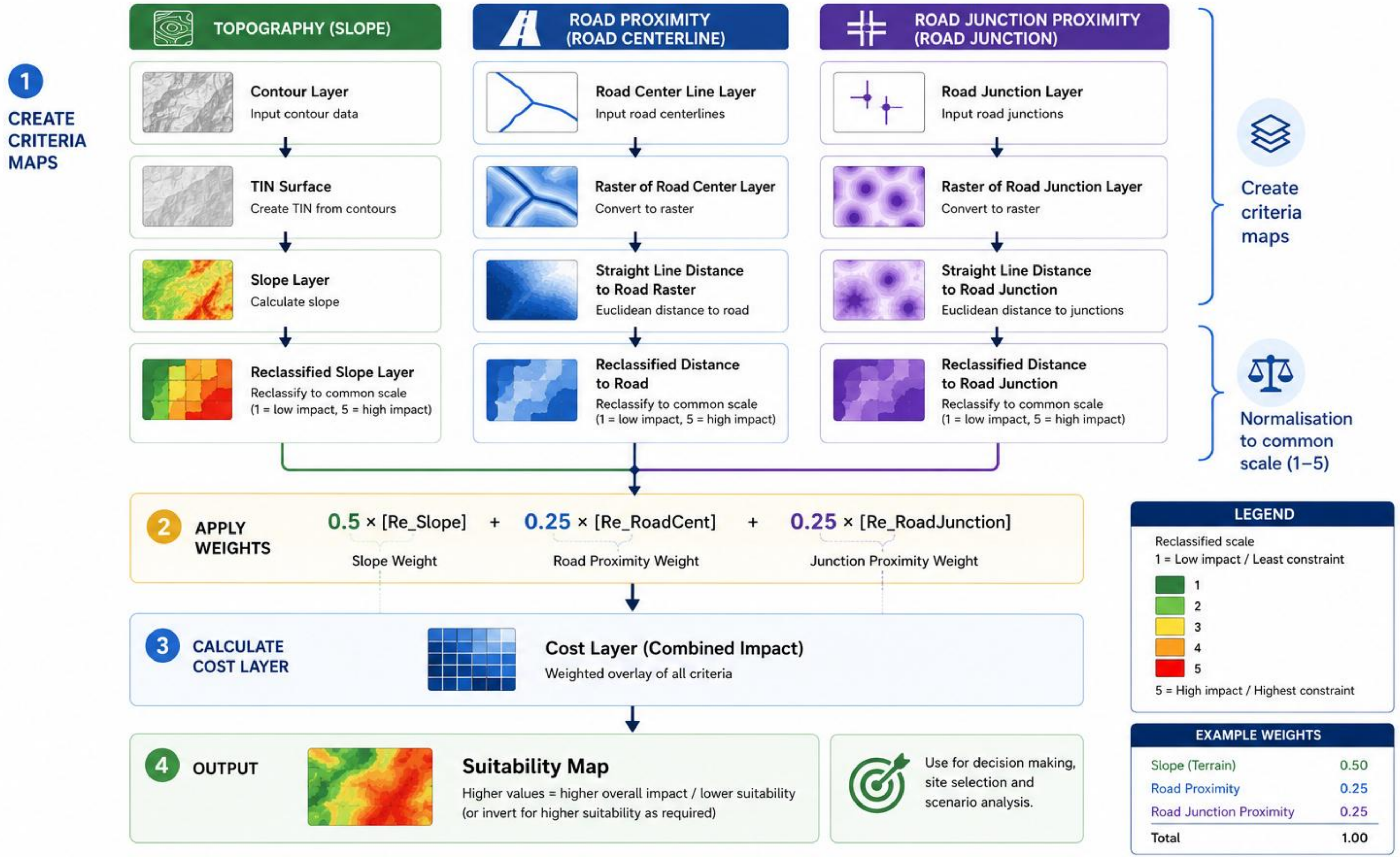
Vary weights to test robustness of the result.

> **MCE provides a systematic, transparent framework for spatial decisions.**

Multi-Criteria Weighted Overlay Workflow

Always ask: 'who chose these weights?'

Create, normalise and combine spatial criteria to produce a suitability map.



KEY CONCEPTS

Today's Takeaways

1. Risk = Hazard × Exposure × Vulnerability — all three must overlap spatially
2. Vulnerability curves convert hazard intensity to expected damage
3. Infrastructure failures cascade — GIS maps the interdependencies
4. MCE: standardise criteria → weight → compute composite → test sensitivity
5. GIS integrates all risk components — the risk map is the ultimate output

NEXT LECTURE

Lecture 11: The Future of GIS - AI Coding Agents

"Design an MCE for selecting a landfill site — what criteria and weights would you use?"

Consider:

Distance from residential areas (buffer analysis)

Groundwater depth and vulnerability

Transport access (distance to major road)

Slope and soil stability

Distance from waterways and sensitive ecosystems

Land cost and ownership

How would you assign weights? What if the community prioritises groundwater protection over transport cost?

Sensitivity Analysis — Testing Weight Choices

What happens when weights change?

Approach 1 — One-at-a-time (OAT):

Vary one weight from 0 to 1, redistribute rest proportionally

Plot: score vs weight for each site — see where rankings swap

Approach 2 — Equal weights baseline:

Compare your weighted result to equal-weight result

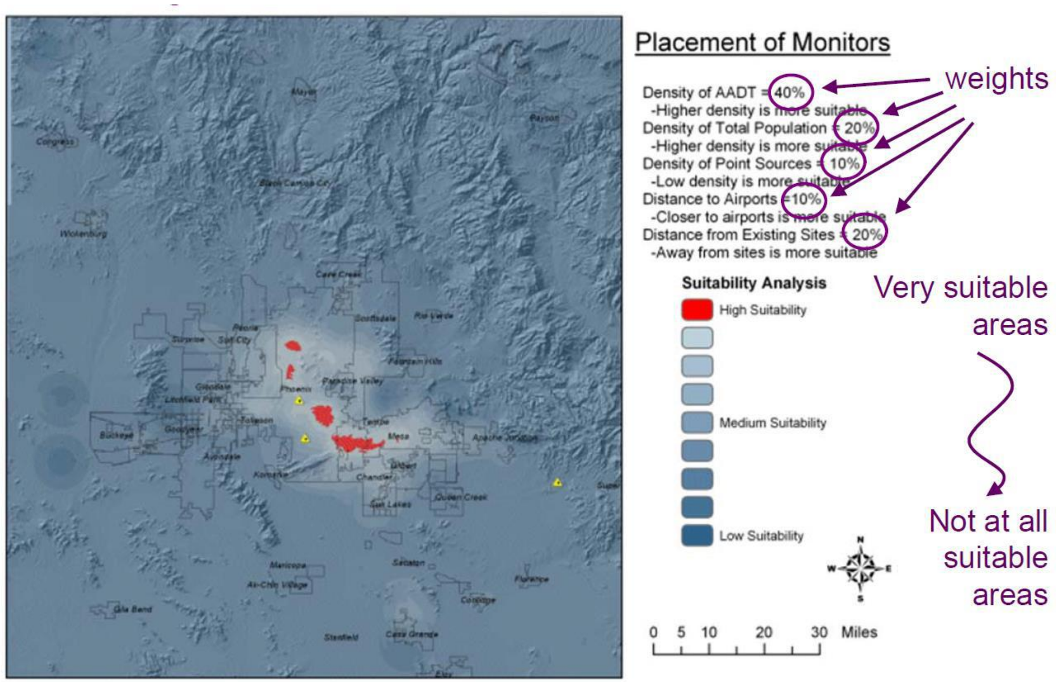
If the same site wins under both, the result is more robust

Approach 3 — Rank reversal check:

At what weight change does the 2nd-best site become 1st?

If a tiny change flips the ranking, the decision is fragile

A robust MCE result survives reasonable weight variation.



MCE applied to air quality monitoring (Phoenix, AZ)

GIS: The Integrating Platform for Risk

Bringing it all together — GIS overlays the three risk components:

Layer 1 — Hazard maps (raster): PGA, flood depth, landslide susceptibility

Layer 2 — Exposure data (vector): buildings, population, infrastructure

Layer 3 — Vulnerability (vector + raster): fragility curves, NZDep, building age

Overlay operations used:

Spatial join — which assets are in the hazard zone?

Raster extract — what is the hazard intensity at each asset?

Zonal statistics — aggregate risk by suburb, district, region

Output: a risk map showing WHERE the highest risk is and WHO is most affected.

Building a Risk Register with GIS

A risk register is a structured database of identified risks:

For each asset (building, bridge, pipe segment):

Location (coordinates, district)

Hazard intensity at that location (from raster extraction)

Vulnerability class (building type, age, condition)

Expected damage (from fragility curve lookup)

Consequence (replacement cost × damage ratio)

Risk score (likelihood × consequence)

GIS automates this for thousands of assets simultaneously.

Output: a GeoDataFrame that IS the risk register — each row is an asset with its risk score.

The Future: GeoAI and Machine Learning

Where GIS is heading:

Automated feature extraction — ML identifies buildings from satellite imagery

Predictive hazard mapping — neural networks learn from past events

Real-time risk monitoring — IoT sensors + GIS dashboards

Digital twins — virtual replicas of cities for scenario testing

Explainable AI — models that tell you WHY a location is high risk

The fundamentals do not change:

You still need to understand data models, CRS, spatial relationships, and risk frameworks.

AI enhances GIS - it does not replace spatial thinking.